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A Novel Hybrid Feature Selection and Ensemble Learning Approach for Mortality Risk Classification in Hepatitis B Patients: An Explainable AI Study

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***Abstract*— Hepatitis B (HBV) is a disease of major health concern that has high rates of fatality and evokes the need to have prognostic tools that will be accurate in providing timely clinical intervention. Current machine learning (ML) predictors of HBV mortality are frequently daunted by method shortcomings, such as un-optimal feature choice and low luminance that do not facilitate their clinical feasibility and reliability. This paper has come up with a holistic framework to surmount these challenges. TrioFS is a newly proposed hybrid feature selection architecture, which involves a sequence combination of deep learning, tree-based, and statistical procedures to determine the most significant predictors. The chosen attributes are applied to train a number of ML models, including several stacking ensemble models, such as a high-accuracy stacking ensemble of K-Nearest Neighbors (KNN) and the Extra Trees Classifier (ETC), with Random Over-Sampling (ROS) applied to address class imbalance. Importantly, we implement Explainable AI (XAI) with SHAP and LIME to make sure that the process of decision-making by the model can be easily explained and made transparent. Our proposed KNN-ETC stacking ensemble performs better than any other work on the benchmark UCI Hepatitis dataset, gaining 98.00 percent accuracy and having perfect ROC-AUC 100.00 percent.**

Keywords: Hepatitis B Mortality Prediction; Stacking Ensemble; Conditional Feature Importance; One-vs-Rest Mutual Information; Clinical Decision Support

I. INTRODUCTION

Hepatitis is an inflammatory liver damage caused most commonly by five hepatotropic viruses, A, B, C, D, and E whose development may lead to the final result represented by fibrosis, cirrhosis, or hepatocellular carcinoma [1]. In 2022, there were 1.3 million viral hepatitis deaths in total (that is, 3,500 deaths a day) with 83% being a result of hepatitis B, 17% a result of hepatitis C and the others only a small percentage. The WHO projections estimate that about 7,134 deaths in 2016 were attributed to hepatitis A, and hepatitis E took away 3,450 lives in 2021, causing 19.47 million acute incidents of this disease in the world; the coinfection with hepatitis D increased the case-fatality rate to 2-20 percent among reported patients of hepatitis B factors, which underlines the magnification potential of hepatitis D in mortality related to hepatitis B. Hepatitis B, such a serious part of this deadly spectrum, could be outlined by its immense human reservoir: there were about 254 million individuals living worldwide with chronic HBV infection in 2022, with nearly half of them between 30 and 54 years old and either undiagnosed or untreated. The control of HBV is the focus of the global hepatitis-elimination efforts, including birth-dose immunization of children worldwide, simplified diagnostics, and an expansion of antiviral therapy [2-5]. In addition to prevention and treatment, proper management of hepatitis B needs to ensure proper prognosis through key biomarkers. Virological responses (HBsAg, HBeAg, anti-HBe, HBV DNA) can measure the infection and replication, whereas hepatocellular damage indicators (ALT, GGT, platelet and FIB-4) act as indices of liver damage. AFP is used to diagnose early hepatocellular carcinoma (HCC) and INR demonstrates coagulopathy. The lifestyle prognosis, along with the epidemiological prognostic factors such as alcohol consumption, obesity, co-infection (HCV, HIV) and vaccination, also plays a role [6]. AI and ML promise to revolutionize care by incorporating a variety of biomarker and clinical data to enhance diagnosis, to predict fibrosis, HCC and survival. These models process complicated files, list important features and improve risk stratification and targeting of interventions [7]. The current research activity is expected to establish powerful prognosis metrics of mortality after the diagnosis of hepatitis B by building on clinical, biochemical, and lifestyle markers to facilitate precise care and promote HBV elimination campaigns.

- TrioFS is a feature selection system, based on a new framework that combines three techniques: deep learning relevance, through TabNet, Conditional Feature Importance (CFI) through Random Forest (RF),

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and One-vs-Rest Mutual Information (OvR-MI) about Classes 0 and 1. Such a novel combination of the neural, statistical and tree methods allows taking the advantage of strengths of each of these and TrioFS is a powerful and efficient mechanism of feature selection.

- In order to assess the survivability prediction of Hepatitis B, three base classifiers- RF and KNN and ETC were trained and stacked into the following stacked ensemble classifiers- RF-KNN, RF-ETC and KNN-ETC, and then Logistic Regression (LR) acts as a meta-learner. The issue of class imbalance was considered through ROS, so the evaluation could be fair.
- The KNN-ETC back-fit had 98% accuracy and a 100% Receiver Operating Characteristic – Area Under the Curve (ROC-AUC). Model reliability was confirmed after internal validation, and to ease the task of decision-making, XAI was used to increase transparency and interpretability.

II. Literature Review

As the number of intelligent health systems has been multiplying, researchers have begun turning to AI models to predict survivability among hepatitis B patients. In this section, important methodologies, findings, and limitations in this field were recounted. Srinivasulu et al. [8] have suggested an optimized Convolutional Neural Network (CNN) model to diagnose hepatitis B with UCI dataset that consists of 155 samples and 19 features. The model was also automatic in its feature extraction and it avoided manual feature selection in order to reduce redundancy and dimensionality. Using 10-fold cross-validation and 70:30 train-test split, it was able to completely outperform previous methods of ML by utilizing evolutionary hyperparameter optimization to achieve 97.50% accuracy. The study however lacked explainability tactics and can therefore not be clinically interpreted. In the same vein, Akbar et al. [9] presented a L1-SVM-AdaBoost which obtained 89.36 accuracy and AUC = 0.649 with the selection of 16 (out of 19) features. Even though it was better than standalone AdaBoost, the model still had low sensitivity (44.44%) because of class imbalance, had no cross-validation, and was not explained formally. Albogamy et al. [10], in turn, introduced a Decision Support System (DSS), based on BiLSTM they used to treat tabular data as sequences and obtained a 95.08% accuracy with oversampling and 10-fold cross-validation. The model performed better in comparison to SVM and CNN but had no feature selection, integration of XAI and reporting of ROC-AUC. The accuracy and F1-score of the two-phase soft voting ensemble by Dutta and Bandyopadhyay [11] that combines Multinomial Naive Bayes (MNB) and RF came to 89.36 percent and 0.89, respectively. Nonetheless, the model did not have feature selection and explainability, ROC-AUC assessment and used a solitary split into 67:33, which can be subject to overfitting as the dataset is not so large. Husain et al. [12] proposed a hybridization with Local Fisher Discriminant Analysis (LFDA) to achieve dimensionality reduction down to the size of the remaining solutions, and an Improved Ant Colony Algorithm (IACA) that optimizes the parameters of Least Squares Support Vector Machine (LSSVM). Their model attained an accuracy level of up to 93.7% and a sensitivity level of 100 percent on the UCI hepatitis dataset. It achieved greater convergence and diagnostic accuracy without any cross-validation or XAI integration. In the same manner, Mishra and Tripathy [13] used a hybrid MLP-GS (Multilayer Perceptron with Genetic Search) model to predict hepatitis risks and extracted 20 features to 9 features via genetic optimization. The model was 82.58 percent accurate and reported RMSE of 0.3908, precision value of 0.814, a recall value of 0.826 and F-measure of 0.818 which is better than standalone MLP. There was however no cross-validation or XAI used.

III. Materials And Methods

This section outlines the data, feature selection, and ML models used to predict Hepatitis B patient survival. It also presents evaluation metrics. **Fig. 1** illustrates the full predictive pipeline to maximize accuracy. The methodological framework starts with data pre-processing, which includes imputation, feature scaling, and label mapping. After an 80/20 train-test split, the training data undergoes two key preparatory steps: applying a novel TrioFS model for feature selection and using ROS to correct for class imbalance. Both base and stacking ensemble models are then trained and optimized through a fully nested hyperparameter tuning process that uses cross-validation with StratifiedKFold and SearchCV. The final stage involves a two-pronged evaluation: a quantitative assessment using performance metrics and a crucial qualitative analysis via XAI to ensure interpretability and transparency.

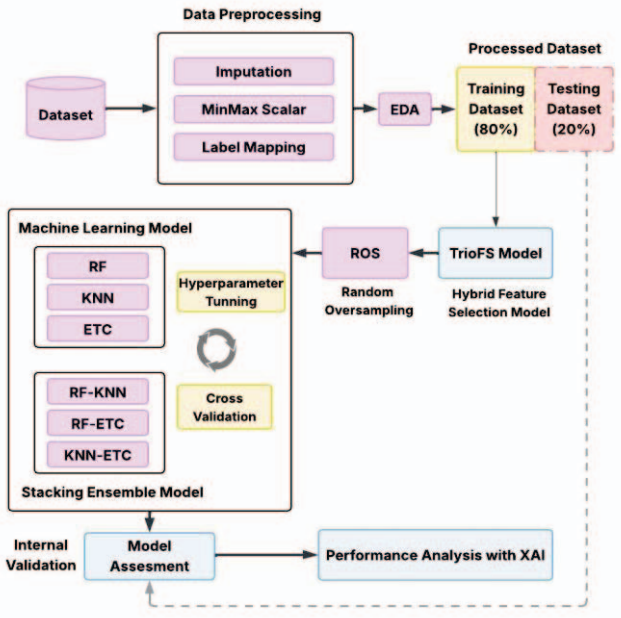


Fig. 1. Overview of the Methodological Framework for Hepatitis B Survivability Prediction

A. Dataset Description

In this paper, the UCI Hepatitis dataset obtained at the UCI ML Repository [14] is a dataset that consists of 155 samples and 20 attributes (six numerical; fourteen categorical). The dependent factor is survival (Live = 1, Die = 2). The dataset should be used to forecast the survival of the patients by informing of clinical, biochemical, and environmental variables.

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Quantitative variables are Bilirubin, Alkaline Phosphatase, SGOT and Albumin, Protime and Age; qualitative ones are Sex, Steroid, Fatigue, Liver status and Treatment history. These are symptoms and therapy records. The information in **TABLE I** on feature types and distribution frequency of classes helped to define fatal factors and render the effective predictive model.

TABLE I. DATASET CHARACTERISTICS

Characteristic	Value
Type	Multivariate
Task	Classification
Instances	155
Features	20
Feature Types	Categorical, Integer, Real
Categorical Features	14
Numerical Features	6
Target Variable	Die/Live
Class Distribution	Imbalanced (123 Live, 32 Die)
Data Source	UCI ML Repository

B. Dataset Cleaning and Preprocessing

Model training and testing was aided by data preprocessing. Examples of missing symptom values in the data set (e.g., fatigue, malaise) were imputed with sklearn SimpleImputer more and numerical data mean. MinMax features scaling brought features to the range [0, 1]. Class was the target variable and was transformed into binary (Die = 0, Live = 1); numerical and categorical variables were transformed into number 1 (True) and 0 (False). The stratified sampling maintained the balance of the classes in the training (80%) and testing (20%) samples. They trained models on a training set and tested them on a test set.

C. Data Examination and Visual Analysis

We have examined the important variables to find patterns and classify the data set better and paid attention to patient age and survival. Among the 155 participants, 79.4 percent (123) survived and 20.6 percent (32) did not. The Density plot, shown in **Fig. 2** illustrates two visualizations, where the survival is concentrated on the ages between 30-50, and there is a possibility of higher risk of mortality with age and therefore the non-surucky level is concentrated on the ages between 40-50. This is further supported by the Count Histogram (right) whereby the younger ones are dominated by the Live group and the older ages assuming a higher place as the Die group.

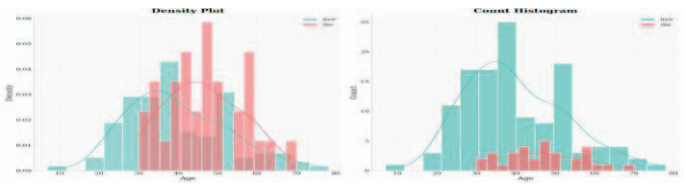


Fig. 2. Age Profile and Analysis of Survivor vs Non-Survivor Patients

D. Hybrid Feature Selection model (TrioFS)

We have designed a hybrid TrioFS as a combination of TabNet, RF-based CFI, and OvR-MI in order to increase the effectiveness of the model. TabNet tracks all the global interactions and CFI and OvR-MI focus on Class 0 (Die) and Class 1 (Live), respectively. The last 15-feature subset is created by overlapping outputs ranked as the same category (top), no more consistent low scorers (steroid, antivirals, anorexia, liver_firm). **Fig. 3** gives the heatmap of normalised scores of all the 19 features and methods. Strict Feature selection was made using training data in order to avoid the leakage and biasedness of the test.

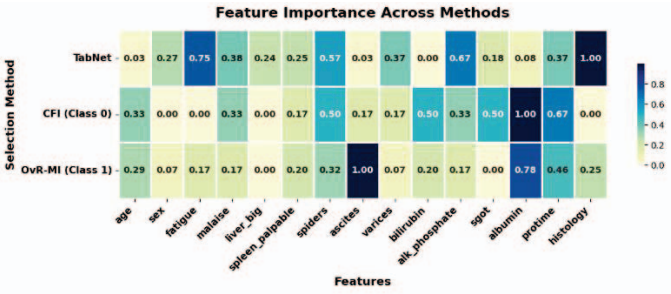


Fig. 3. Normalized Feature Importance Scores Across TrioFS Component Models.

The TrioFS design that will be proposed is a so-called hybrid feature selection strategy, which combines deep learning, tree-based, and statistical approaches. TrioFS protects against bias by contrasting with single-method selectors in that it brings together insights into individual classes using CFI and OvR-MI and the inter-feature learning capacity of TabNet. This allows selecting both world relevant and class relevant features with coverage of broad patterns with comparatively smaller feature set. **Algorithm 1** step-by-step describes the entire selection process.

Algorithm 1. TrioFS Model: Hybrid Multi-Method Feature Selection

Input: Dataset D = {X, y}, where X ∈ $\mathbb{R}^{nxd}$, y ∈ {0, 1, ..., c-1}
Output: Final selected feature subset F_final ⊆ {1, 2, ..., d}
Parameters: n_features_tabnet, cfi_threshold, ovr_mi_threshold
Initialize
Split (X, y) into train/test (80/20)
Train TabNet classifier on (X_train, y_train)
Extract feature importances I_TabNet
if n_features_tabnet specified then
    F_TabNet ← top-k features from I_TabNet
else
    F_TabNet ← {i : I_TabNet[i] > mean(I_TabNet)}
end if
y_binary ← (y == 0)
Train RF on (X_train, y_binary_train)
baseline ← accuracy (RF, X_test, y_binary_test)
for each feature i do
    X_perm ← X_test with feature i permuted
    CFI[i] ← max (0, baseline - accuracy (RF, X_perm, y_binary_test))
end for
F_CFI ← {i : CFI[i] ≥ max(cfi_threshold, mean(CFI))}
y_binary ← (y == 1)
MI ← mutual_info_classif(X, y_binary)
F_OvR_MI ← {i : MI[i] ≥ max(ovr_mi_threshold, mean(MI))}
F_final ← F_TabNet ∪ F_CFI ∪ F_OvR_MI
Sort F_final by index
return F_final

E. Class Imbalance Handling

To correct class imbalance in the training data (98 'Live' vs. 26 'Die' instances), we used Random Over-Sampling (ROS). In

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this manner, the technique balanced the dataset by repeating samples from the minority ('Die') class. ROS is applied only to the training set to prevent information leakage, resulting in our balanced intentional subset of 98 data points per class, as shown in **Fig. 4**. This is an important step in the process of creating a non-partial, proper and efficacious model with proper balancing.

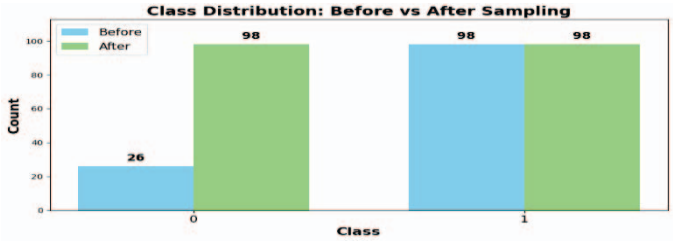


Fig. 4. Class Distribution Before and After Applying ROS

F. Machine Learning and Stacking Ensemble Models

The paper evaluates three base classifiers (RF, KNN, ETC) with three heterogeneous stacking ensembles (RF-ETC, KNN-ETC, KNN-RF) using Logistic Regression as a meta-learner. We developed the model on a training dataset (80%) via nested cross-validation methods. A stratified 10-fold Stratified K-Fold was used for the outer loop. This method is based on the principle shown in (Equation 1), which defines $p(c_j)$ as the percentage of a category c_j within the dataset S. This value, derived from the count of samples in that class, $|Sc_j|$ ensures that each fold, F_i , preserves the original class distribution, a methodological necessity to minimize variance in performance estimates and allow an accurate assessment of model performance.

$$p(c_j) = \frac{|Sc_j|}{|S|} \quad (1)$$

Inside each of the outer folds, we used RandomizedSearchCV to optimise hyperparameters in an inner validation loop. This process is generally summarized as determining the hyperparameter vector θ* which maximizes the mean cross-validation score (CV_score) of a fixed number of sampled parameter configurations (θ_i), as shown in (Equation 2).

$$\theta^* = \arg \max_{\theta_i \epsilon \{\theta_1, \ldots, \theta_n\}} CV_score(\theta_i) \quad (2)$$

The parameter grids used in this search were intentionally constructed as such that for base learners, 5:3 hyperparameter-to-range ratio were used, yielding 162 unique sets each. For the stacking ensembles specifically, the Logistic Regression meta-learner was additionally tuned using a separate 3:3 grid structure for its regularization parameters, resulting in 27 configurations.

G. Performance Metrics

For the evaluation of the model, fundamental metrics derived from the confusion matrix were employed, including Accuracy, Precision, Recall (Sensitivity), and F1-Score [15]. In addition, to obtain an overall measure of the discriminative power of each model at all classification thresholds, ROC-AUC was also computed .

IV. Results And Discussion

TABLE II lists the performance of all models on an independent test set. By looking at the performance of all base classifiers, RF and ETC had performed competitive performances with an accuracy = 96.00% and ROC-AUC = 100.00%. The KNN model is also quite resilient, but slightly outperformed by the tree ensembles. Based on the evaluation of stacked models, we can conclude that KNN-ETC is our best model. It obtained an accuracy, precision and recall of 98.00%, F1-Score of 98.00% and perfect ROC-AUC of 100% . Which is making it the best model among this evaluation table.

TABLE II. PERFORMANCE EVALUATION OF CLASSIFIERS

Model Name	Accuracy	Precision	Recall	F1 Score	ROC-AUC
RF	96.00%	96.30%	96.00%	95.99%	100.00%
KNN	94.00%	94.64%	94.00%	93.98%	98.00%
ETC	96.00%	96.30%	96.00%	95.99%	100.00%
RF-ETC	96.00%	96.30%	96.00%	95.99%	100.00%
KNN-ETC	98.00%	98.08%	98.00%	98.00%	100.00%
KNN-RF	96.00%	96.30%	96.00%	95.99%	100.00%

Best Performing Model:

Stacking ensemble with KNN and ETC was found to be the best performing and more stable model. The confusion matrices (**Fig. 5**) have done a great job with almost perfect classification from the test set with only a single False Positive and zero False Negatives. This result is underscored by the comparative ROC analysis in **Fig. 6**, which shows the KNN-ETC model demonstrating flawless discriminative power with a perfect test AUC of 1.0000, solidifying its status as the premier model in this study. This also confirms the stability of the model, as evident from the small standard deviation across all 10 folds shown in **TABLE III**.

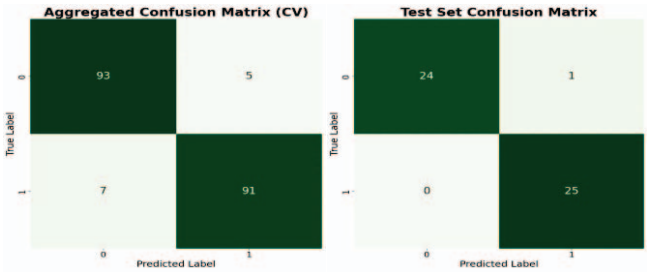


Fig. 5.KNN-ETC Confusion Matrices: 10-Fold CV and Test Set

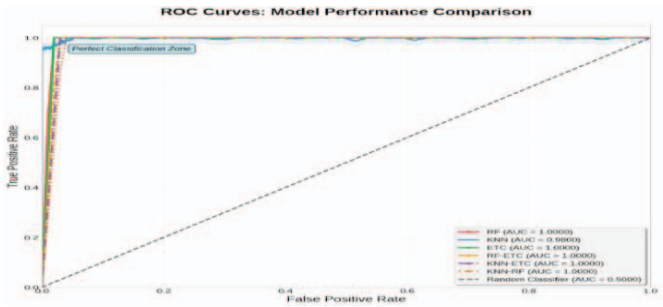


Fig. 6. ROC Curve Comparison Between All Models

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TABLE III. METRICS ACROSS 10 FOLDS FOR THE KNN-ETC MODEL

K-Fold	Accuracy	Precision	Recall	F1-Score
Fold 1	1	1	1	1
Fold 2	0.9	0.9167	0.9	0.899
Fold 3	0.95	0.9545	0.95	0.9499
Fold 4	0.85	0.8535	0.85	0.8496
Fold 5	0.95	0.9545	0.95	0.9499
Fold 6	0.95	0.9545	0.95	0.9499
Fold 7	0.9474	0.9526	0.9474	0.9474
Fold 8	0.9474	0.9526	0.9474	0.9474
Fold 9	0.9474	0.9526	0.9474	0.9474
Fold 10	0.9474	0.9526	0.9474	0.9474
Mean	0.9389	0.9444	0.9389	0.9388
Standard Deviation	0.0372	0.0356	0.0372	0.0373
95% Confidence Interval	±0.0231	±0.0221	±0.0231	±0.0231

This model is successful due to the synergy between its distinct base learners, who's optimized hyperparameters are shown in **TABLE IV**. The KNN part is good for local, instance-based pattern recognition and the ETC part captures more global, non-linear relationships. The LR meta-learner with fitted L2 regularization utilizes these attributions to combine these complementary predictions into a composite final model that generalizes better.

TABLE IV. FINAL HYPERPARAMETER CONFIGURATION OF THE BEST PERFORMING MODEL (KNN-ETC)

Model	Hyperparameter	Best Value
KNN	n_neighbors	5
	weights	distance'
	algorithm	kd_tree'
	p (Minkowski Power)	3
	leaf_size	40
ETC	n_estimators	150
	max_depth	12
	min_samples_split	2
	bootstrap	TRUE
	min_samples_leaf	1
Logistic Regression (Meta-Learner)	c (Inverse Regularization)	0.01
	penalty	l2'
	solver	liblinear'

Performance Analysis with Explainable AI:

For transparency, we interpreted SHAP and LIME predictions for the KNN-ETC model, with the SHAP summary plot (**Fig. 7**) highlighting histology, malaise and ascites as the top predictors of Hepatitis B mortality. Local, instance-specific explanations, visualized in the SHAP force plot (**Fig. 8**) and LIME outputs (**Figs. 9 and 10**), confirmed that these same features consistently drove individual predictions. **Fig. 8** presents SHAP force plot used to visualize the drivers of a single prediction as how the features like bilirubin and age increase cumulatively to drive the output of the model all the way to the final prediction. Moreover, LIME presents the causes of individual predictions categorically. Among the predictors of survival (**Fig. 9**), not having severe symptoms, i.e., spiders <= 0.00, ascites <= 0.00, and malaise <= 0.00 were on the top of the list. On the other hand, in the case study of one patient who is predicted to die (**Fig. 10**), the prediction of the model depended on the presence of the same signs of advanced liver disease, such as spiders, fatigue, and a palpable spleen . Its reliance on such features, of established prognostic marker for major liver decompensation such as ascites and elevated bilirubin level [16][17] and biopsy findings (histology) [18], gives a clinical justification for its decision support, along with reinforcing the trustworthiness of the model.

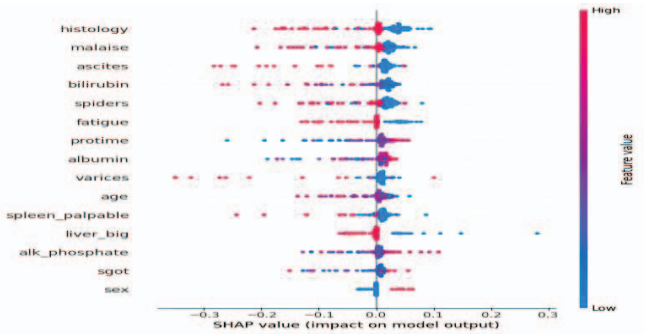


Fig. 7: SHAP Summary: KNN-ETC Feature Importance

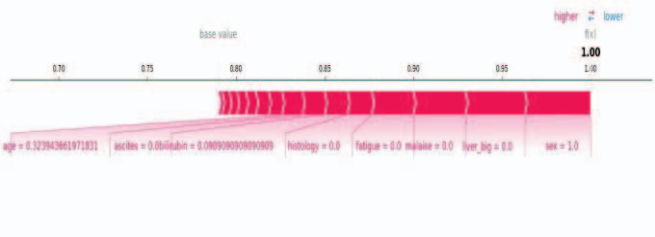


Fig. 8: SHAP Force Plot: Single Prediction Feature Impact



Fig. 9: LIME Explanation: Sample Prediction (Live)

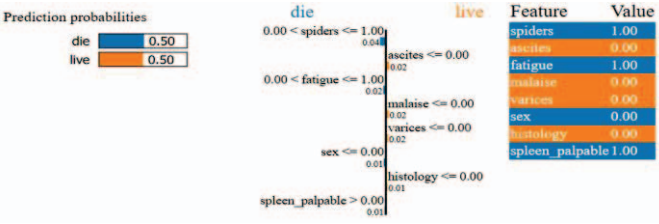


Fig. 10: LIME Explanation: Sample Prediction (Die)

Comparison with Other Related Works :

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We present empirical benchmarking results in **TABLE V**, which exhibit our proposed KNN-ETC ensemble (98% accuracy, 100% ROC-AUC) significantly outperforming other models on the UCI Hepatitis dataset compared to the highest reported accuracy result of 97.50% by Srinivasyulu et al [8]. Our framework pioneers critical methodologies including the novel TrioFS feature selection model, robust 10-fold cross-validation, and most importantly, XAI to achieve interpretability.

TABLE V. PROPOSED MODEL COMPARISON WITH RELATED WORKS

Existing Literatures	Dataset	Classification Method	Accuracy	ROC-AUC	Feature Selection Model	Cross Validation	XAI
Srinivasulu et al. [8]	UCI Hepatitis	CNN	97.50%	-	✕	✓	✕
Akbar et al. [9]	UCI Hepatitis	AdaBoost	89.36%	64.90 %	L1-SVM	✕	✕
Albogamy et al. [10]	UCI Hepatitis	DSS	95.08%	-	✕	✓	✕
Dutta and Bandyopadhyay [11]	UCI Hepatitis	Soft Voting with RF-MNB	89.36%	-	✕	✕	✕
Husain et al. [12]	UCI Hepatitis	IACA-LSSVM on LFDA	93.70%	-	✕	✕	✕
Mishra and Tripathy [13]	UCI Hepatitis	MLP-GS	82.58%	-	GS	✕	✕
Proposed Model	**UCI Hepatitis**	**Ensemble Stacking with KNN–ETC**	**98%**	**100%**	**TrioFS Model**	✓	✓

V. CONCLUSION

This paper proposed a powerful, multidimensional model, which is able to predict mortality among Hepatitis B patients and it helps to fill crucial gaps in precision, research design and explanation that previous studies have encountered in a variety of directions. This paper also introduced the TrioFS hybrid feature selection model, a nested cross-validation process for stacking ensembles, and a novel implementation of XAI for transparency. Experimental results validate our method and show that we obtain a new benchmark accuracy of 98% and ROC-AUC of 100% on the UCI dataset by using KNN-ETC stacking ensemble, outperforming all benchmarked models. XAI further confirmed its clinical significance by revealing the dependence of the model on known prognostic markers. The consistency of the statistical performance with the clinical knowledge corroborates the model by identifying it as reliable and valid. This paper indicates future research directions in a number of distinct areas. Next steps include external validation on a diverse dataset and consideration of LLMs to generate features from clinical notes. The study lays a solid ground for developing rapid, predictive, and explainable AI models for clinical hepatology.

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